Numerical Range and Cartesian Decomposition Enclosures for Zeros of Quaternionic Polynomials

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October 11, 2026

Abstract

Traditional zero bounds for polynomials with structured or monotonic coefficients—from classical Cauchy and Kojima bounds to the Eneström-Kakeya theorem—all rely on a simple compromise: they trap roots inside generic, concentric spherical balls centered at the origin. Doing this throws away the natural directional asymmetry between the real axis and the imaginary subspace. In this paper, I break away from the spherical system by extracting coordinate-decomposed bounds directly from the quaternionic numerical range $W(C)$ of the direct Frobenius companion matrix $C \in M_n(\mathbb{H})$.

By evaluating the Cartesian decomposition $C = H + K$, where $H = \frac{1}{2}(C + C^*)$ is quaternionic Hermitian and $K = \frac{1}{2}(C - C^*)$ is skew-Hermitian, I extend the classical Bendixson-Hirsch bounds to the non-commutative quaternionic setting using Rayleigh quotients on $\mathbb{H}^n$. The Hermitian component acts as an in-line stretching engine that measures the real coordinate of every zero, while the skew-Hermitian component acts as a rotational engine that measures the imaginary magnitude. Every right zero $q \in \mathbb{H}$ of any monic polynomial over $\mathbb{H}$ unconditionally satisfies $\lambda_{\min}(H) \leq \mathrm{Re}(q) \leq \lambda_{\max}(H)$ and $|\,\mathrm{Im}(q)| \leq \sigma_{\max}(K)$, carving out an orthogonal cylindrical enclosure $\mathcal{B}_{\mathrm{Cart}}$ in $\mathbb{R}^4$: a 1D real coordinate slab intersected with a 3D Euclidean imaginary ball.

For polynomials with real coefficients that satisfy Eneström-Kakeya monotonicity ($0 < a_0 \leq \cdots \leq a_n = 1$), all zeros lie within the unit ball $\mathbb{B}_1$. Slicing $\mathbb{B}_1$ with the Cartesian cylinder $\mathcal{B}_{\mathrm{Cart}}$ produces an anisotropic enclosure $\Omega = \mathbb{B}_1 \cap \mathcal{B}_{\mathrm{Cart}}$ that shears off empty volume along the real axis and imaginary radius. For degree $n = 2$, this eliminates a mean 2D area of 60.57% and a mean 4D hypervolume of 76.31% relative to $\mathbb{B}_1$ (averaging 24.72% in 2D and 26.72% in 4D at $n = 3$). For polynomials with general non-real quaternionic coefficients where zeros can exceed $|q| = 1$ (rendering $\mathbb{B}_1$ inapplicable), $\mathcal{B}_{\mathrm{Cart}}$ unconditionally traps the full 4D spectrum.

Furthermore, using Cauchy's interlacing theorem on the leading $(n-1) \times (n-1)$ Toeplitz blocks of H and K, I prove that the removable 2D area fraction over $\mathbb{B}_1$ is bounded above by $\frac{8\pi^2}{3n^3} + \mathcal{O}(n^{-5})$ and the 4D volume fraction by $\frac{16\pi^2}{3n^3} + \mathcal{O}(n^{-5})$ (governed by the 3D equatorial collar of the imaginary cutoff). This establishes a rigorous upper bound: geometric improvements over $\mathbb{B}_1$ decay at rate $\mathcal{O}(n^{-3})$ across all dimensions, while the empirical growth of $\lambda_{\max}(H) > 1$ at $n \geq 8$ terminates active truncation.

Finally, I provide a formal mathematical post-mortem of alternative row-sum heuristics, proving why full Cassini unions collapse to the unit disk (0% savings for $n \geq 3$) and why isolating a single Cassini oval leaks zeros (23.67% failure rate). At first, this was my approach to the problem, but it was in vain. I decided to include this post-mortem in the paper so anyone exploring this path can avoid making the exact same mistake. Numerical stability is validated across an adversarial stress test of 196,059 polynomials spanning seven test batteries, yielding zero containment violations.

Keywords: Quaternionic polynomials, Zero bounds, Numerical range, Companion matrix, Cartesian decomposition, Bendixson-Hirsch bounds, Eneström-Kakeya theorem, Cauchy interlacing.

AMS Subject Classification (2020): 12E15, 15A18, 15A60, 15B57, 30C15.

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1 Introduction

Quaternions ($\mathbb{H}$) are the natural language for non-commutative spatial math, powering 3D computer graphics, spatial robotics, rigid body dynamics, control engineering, and quantum mechanics [1–3]. A quaternion $q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$ has one real scalar coordinate $\mathrm{Re}(q) = q_0$ and a three-dimensional imaginary vector $\mathrm{Im}(q) = q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$, governed by Hamilton's relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$, with conjugate $q^* = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k}$ and Euclidean norm $|q| = \sqrt{qq^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}$.

A fundamental problem in hypercomplex analysis is zero localization: trapping the roots of a quaternionic polynomial inside tight, reliable boundaries without having to run heavy, iterative root-finding algorithms. For complex polynomials, the classical Eneström-Kakeya theorem guarantees that if $P(z) = \sum_{\nu=0}^{n} a_\nu z^\nu$ has positive monotonic real coefficients $0 < a_0 \leq a_1 \leq \cdots \leq a_n = 1$, all zeros sit strictly inside the closed unit disk $|z| \leq 1$. Several modern generalizations have attempted to push this result to quaternions with restricted coefficients [4–6].

Isotropic Baselines and the Spherical Compromise. Standard zero bounds for monic polynomials $T(q) = \sum_{\nu=0}^{n-1} a_\nu q^\nu + q^n$ include:

1. **Cauchy's Bound:** All zeros satisfy $|q| \leq 1 + \max_{0\leq\nu\leq n-1} |a_\nu|$. Under monotonicity $a_\nu \leq 1$, this guarantees $|q| \leq 2$.
2. **Kojima's Bound:** All zeros satisfy $|q| \leq \max(|a_0|, 1 + |a_1|, \ldots, 1 + |a_{n-1}|) \leq 2$.
3. **Fujiwara's Bound:** All zeros satisfy $|q| \leq 2\max_{1\leq j\leq n} |a_{n-j}|^{1/j}$.
4. **Eneström-Kakeya Bound:** When coefficients are real and positive monotonic ($0 < a_0 \leq \cdots \leq a_n = 1$), all zeros satisfy $|q| \leq 1$.

All of these classical benchmarks share a major flaw: they are strictly *isotropic*. They draw giant spheres $\mathbb{B}_R = \{q \in \mathbb{H} : |q| \leq R\}$ centered at the origin, treating every single direction in 4D space as identical. But polynomials with structured or monotonic coefficients have roots that are naturally asymmetric between the real axis and the imaginary subspace. Drawing a giant sphere is a blunt, over-bounding method that leaves massive amounts of empty space inside, forcing us to spend unnecessary compute on search regions. While Bendixson-Hirsch bounds and numerical range localization (field of values $W(C)$) on Frobenius companion matrices are well studied in complex matrix theory [7,8], formulating them over the non-commutative algebra $\mathbb{H}$ and studying their geometric interaction with Eneström-Kakeya monotonicity has never been systematically explored for this problem.

Motivation: Structural Failure of Row-Sum and Cassini Heuristics. One might assume that applying Brauer's Cassini oval theorem to the companion matrix would fix this issue. But as I personally tested, it simply doesn't work:

1. **Unit-Disk Collapse of Full Unions:** If you keep all row pairs as mathematically required by Brauer's theorem, the intermediate rows of the Frobenius companion matrix (which all have center 0 and row sum 1) force the full Brauer union to cover the unit disk $\mathbb{D}(0, 1)$ identically for $n \geq 3$, yielding exactly 0% area reduction.
2. **Root Leakage in Isolated Ovals:** If, to force a tighter boundary, you isolate the terminal-row Cassini oval, containment gets completely destroyed. Real negative zeros escape the isolated oval whenever their magnitude exceeds a_{n-1}, producing a 23.67% empirical failure rate across the 10,000 test polynomials I ran.

Seeing these row-sum methods fail so badly pushed me to look for a completely different approach. That is how I arrived at an operator-theoretic framework based on the quaternionic numerical range.

The Numerical Range and Cartesian Decomposition Approach. To completely bypass the limitations of row-sum norm bounds, I attack the problem using the **quaternionic numerical range** of the direct companion matrix:

$$W(C) = \{x^* C x : x \in \mathbb{H}^n,\ \|x\|_2 = 1\} \subset \mathbb{H}. \tag{1}$$

Instead of treating the matrix as an indivisible black box, I split it into its Cartesian components $C = H + K$. The Hermitian component $H = \frac{1}{2}(C + C^*)$ acts as an in-line stretching engine that captures the real coordinate of every zero, while the skew-Hermitian component $K = \frac{1}{2}(C - C^*)$ acts as a rotational engine that captures the imaginary magnitude. By evaluating Rayleigh quotients on $\mathbb{H}^n$, I derive coordinate-decomposed Bendixson-Hirsch bounds on $W(C)$:

$$\mathrm{Re}(W(C)) \subseteq [\lambda_{\min}(H), \lambda_{\max}(H)], \quad |\,\mathrm{Im}(W(C))| \subseteq [0, \sigma_{\max}(K)].$$

Because every right zero of any monic polynomial is a right eigenvalue of C, every zero unconditionally lands inside $W(C)$. For real monotonic polynomials, intersecting this cylindrical enclosure $\mathcal{B}_{\mathrm{Cart}}$ with the Eneström-Kakeya unit ball $\mathbb{B}_1$ produces an anisotropic enclosure $\Omega = \mathbb{B}_1 \cap \mathcal{B}_{\mathrm{Cart}}$ that:

1. **Guarantees containment:** No roots can mathematically escape the enclosure (100% containment across all monic polynomials over $\mathbb{H}$, confirmed across all my tests).
2. **Delivers massive geometric reductions for low degrees:** For $n = 2$, it eliminates an average of 60.57% of the 2D cross-sectional area and 76.31% of the 4D hypervolume relative to the classical unit ball (averaging 24.72% in 2D and 26.72% in 4D at $n = 3$).
3. **Establishes a closed-form asymptotic upper bound:** Because the leading $(n-1) \times (n-1)$ submatrices of H and K are tridiagonal Toeplitz blocks, Cauchy's interlacing theorem proves that the removable 2D area fraction over $\mathbb{B}_1$ is strictly bounded above by $\frac{8\pi^2}{3n^3} + \mathcal{O}(n^{-5})$ and the 4D volume fraction by $\frac{16\pi^2}{3n^3} + \mathcal{O}(n^{-5})$. This proves that geometric gains decay at rate $\mathcal{O}(n^{-3})$ in both 2D and 4D, while the empirical growth of $\lambda_{\max}(H) > 1$ at $n \geq 8$ terminates active truncation.

The paper is structured as follows. Section 2 formalizes the numerical range Cartesian enclosure and proves the Rayleigh-quotient bounding theorem. Section 3 breaks down the geometric area/volume and proves the Cauchy interlacing cap shrinkage theorem. Section 4 provides a formal post-mortem of why row-sum and Cassini oval baselines collapse. Section 5 reports empirical stress-test results across 196,059 adversarial polynomials. Concluding remarks follow in Section 6.

2 The Numerical Range and Cartesian Decomposition Framework

Consider a monic left quaternionic polynomial of degree n:

$$T(q) = a_0 + a_1 q + a_2 q^2 + \cdots + a_{n-1} q^{n-1} + q^n, \quad a_\nu \in \mathbb{H}. \tag{2}$$

A quaternion $\lambda \in \mathbb{H}$ is a right zero of $T(q)$ if $T(\lambda) = 0$.

2.1 Companion Matrix and Symplectic Embedding

The zeros of $T(q)$ coincide with the right eigenvalues of its $n \times n$ direct Frobenius companion matrix $C \in M_n(\mathbb{H})$:

$$C = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{pmatrix}. \tag{3}$$

Specifically, if $x = (x_1, \dots, x_n)^T \in \mathbb{H}^n \setminus \{0\}$ satisfies $Cx = x\lambda$, eliminating coordinates yields $x_k = x_1\lambda^{k-1}$ for $k = 1, \dots, n$, and the terminal row enforces $T(\lambda) = 0$ (taking $x_1 = 1$).

I decompose C into its quaternionic Cartesian components:

$$C = H + K, \quad \text{with } H = \frac{C + C^*}{2}, \quad K = \frac{C - C^*}{2}, \tag{4}$$

where $C^* = (\bar{c}_{ji})$ is the quaternionic conjugate transpose. By construction, $H^* = H$ is quaternionic Hermitian, and $K^* = -K$ is quaternionic skew-Hermitian.

To evaluate spectra over the non-commutative division algebra $\mathbb{H}$, I use the symplectic embedding $\chi : \mathbb{H} \to M_2(\mathbb{C})$ [2]:

$$\chi(q) = \chi(q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}) = \begin{pmatrix} q_0 + \mathrm{i}q_1 & q_2 + \mathrm{i}q_3 \\ -q_2 + \mathrm{i}q_3 & q_0 - \mathrm{i}q_1 \end{pmatrix}, \tag{5}$$

where $\mathrm{i} = \sqrt{-1} \in \mathbb{C}$ denotes the complex imaginary unit, distinguished from the quaternionic units $\mathbf{i}, \mathbf{j}, \mathbf{k}$. Applying χ entrywise lifts any matrix $A \in M_n(\mathbb{H})$ to $\chi(A) \in M_{2n}(\mathbb{C})$. The mapping χ is an injective $*$-algebra homomorphism into $M_{2n}(\mathbb{C})$:

$$\chi(AB) = \chi(A)\chi(B), \quad \chi(A^*) = (\chi(A))^*.$$

As a consequence:

1. $\chi(H)$ is a complex Hermitian matrix of size $2n \times 2n$. Its $2n$ eigenvalues are all real and appear in pairs of equal value with multiplicity 2.

2. $\chi(K)$ is complex skew-Hermitian, so its eigenvalues are purely imaginary: $\pm\mathrm{i}\mu_k$ ($k = 1, \dots, n$). Consequently, $-\mathrm{i}\chi(K)$ is a complex Hermitian matrix whose real eigenvalues appear in opposite-sign pairs $\pm\mu_k$ (n pairs in total), corresponding to the singular values $\mu_k \geq 0$ of $\chi(K)$.

2.2 Numerical Range Rayleigh-Quotient Bounding Theorem

Theorem 2.1 (**Quaternionic Numerical Range Cartesian Enclosure**). *Let $T(q) = \sum_{\nu=0}^{n} a_\nu q^\nu$ be any monic quaternionic polynomial ($a_n = 1$). Let $C \in M_n(\mathbb{H})$ be its direct companion matrix, with Cartesian components $H = \frac{1}{2}(C + C^*)$ and $K = \frac{1}{2}(C - C^*)$. Then every right zero $\lambda \in \mathbb{H}$ of $T(q)$ satisfies:*

$$\lambda_{\min}(H) \leq \mathrm{Re}(\lambda) \leq \lambda_{\max}(H), \tag{6}$$

and

$$|\,\mathrm{Im}(\lambda)| \leq \sigma_{\max}(K) = \lambda_{\max}(-\mathrm{i}\chi(K)), \tag{7}$$

where $\lambda_{\min}(H)$ and $\lambda_{\max}(H)$ denote the minimum and maximum eigenvalues of $\chi(H)$, and $\sigma_{\max}(K)$ is the largest singular value of $\chi(K)$.

Proof. Let $\lambda \in \mathbb{H}$ be a right zero of $T(q)$. Then λ is a right eigenvalue of C, meaning there exists a non-zero eigenvector $x \in \mathbb{H}^n$ such that $Cx = x\lambda$. Normalizing x so that $\|x\|_2^2 = x^*x = 1$, the Rayleigh quotient of C evaluates to:

$$x^*Cx = x^*(x\lambda) = (x^*x)\lambda = \lambda \in W(C).$$

Taking the quaternionic conjugate transpose yields $(x^*Cx)^* = x^*C^*x = \lambda^*$. Evaluating the Rayleigh quotient of the Hermitian component $H = \frac{1}{2}(C + C^*)$:

$$x^*Hx = \frac{1}{2}\left(x^*Cx + x^*C^*x\right) = \frac{1}{2}(\lambda + \lambda^*) = \mathrm{Re}(\lambda).$$

By standard Rayleigh quotient bounds on quaternionic Hermitian matrices (Zhang [2, Section 6]), $\lambda_{\min}(H) \le x^*Hx \le \lambda_{\max}(H)$, which proves Eq. (6).

Similarly, for the skew-Hermitian component $K = \frac{1}{2}(C - C^*)$:

$$x^*Kx = \frac{1}{2}\left(x^*Cx - x^*C^*x\right) = \frac{1}{2}(\lambda - \lambda^*) = \mathrm{Im}(\lambda).$$

The Euclidean norm satisfies $|\,\mathrm{Im}(\lambda)| = |x^*Kx| \le \|K\|_2 = \|\chi(K)\|_2 = \sigma_{\max}(\chi(K))$. Since $-\mathrm{i}\chi(K)$ is Hermitian, its singular values coincide with the absolute values of its eigenvalues:

$$|\,\mathrm{Im}(\lambda)| \le \|K\|_2 = \lambda_{\max}(-\mathrm{i}\chi(K)).$$

This completes the proof. □

2.3 The Enclosure Regions $\mathcal{B}_{\mathbf{Cart}}$ and Ω

In 4D Euclidean space $\mathbb{R}^4 \cong \mathbb{H}$, these spectral bounds define an orthogonal hyper-cylinder:

$$\mathcal{B}_{\mathrm{Cart}} = \{q \in \mathbb{H} : \lambda_{\min}(H) \le \mathrm{Re}(q) \le \lambda_{\max}(H),\ \|\,\mathrm{Im}(q)\|_2 \le \sigma_{\max}(K)\}\,. \tag{8}$$

Geometrically, $\mathcal{B}_{\mathrm{Cart}}$ is the Cartesian product of the 1D real spectral interval $[\lambda_{\min}(H), \lambda_{\max}(H)]$ with the 3D Euclidean imaginary ball $\mathbb{B}_3(0, \sigma_{\max}(K))$. While 2D planar slices ($\mathbb{C} \subset \mathbb{H}$) look like rectangular bounding boxes, in 4D hypervolume the region is actually cylindrical, possessing continuous rotational symmetry across the imaginary units $\mathbf{i}, \mathbf{j}, \mathbf{k}$. Theorem 2.1 guarantees that $\mathcal{B}_{\mathrm{Cart}}$ traps all zeros of **any** monic quaternionic polynomial unconditionally.

When the polynomial coefficients are real and satisfy the classical Eneström-Kakeya condition $0 < a_0 \le a_1 \le \cdots \le a_n = 1$, all zeros are additionally guaranteed to lie inside the unit ball $\mathbb{B}_1 = \{q \in \mathbb{H} : |q| \le 1\}$. In this case, I define the composite enclosure:

$$\Omega = \mathbb{B}_1 \cap \mathcal{B}_{\mathrm{Cart}}. \tag{9}$$

For general quaternionic polynomials with non-real coefficients, zeros can exceed $|q| = 1$ (for example, $T(q) = q^2 + q + (1 + 0.15\mathbf{i})$ has a zero with $|q| \approx 1.049$). In those cases, the unit ball $\mathbb{B}_1$ cannot be used, and $\mathcal{B}_{\mathrm{Cart}}$ alone serves as the unconditional enclosure.

2D Cross-Sectional Area and 4D Hypervolume Definitions.

1. **2D Planar Cross-Section ($\mathbb{C} \subset \mathbb{H}$):** With coordinates $x = \mathrm{Re}(q)$ and $y = |\,\mathrm{Im}(q)|$, setting $x_{\mathrm{left}} = \max(-1, \lambda_{\min}(H))$ and $x_{\mathrm{right}} = \min(1, \lambda_{\max}(H))$, the cross-section is:

$$\Omega_{2\mathrm{D}} = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1,\ x_{\mathrm{left}} \le x \le x_{\mathrm{right}},\ |y| \le \sigma_{\max}(K)\right\}. \tag{10}$$

   Defining the boundary profile $y_{\mathrm{top}}(x) = \min\left(\sqrt{\max(0, 1 - x^2)},\ \sigma_{\max}(K)\right)$, the 2D area evaluates to:

$$\mathrm{Area}(\Omega_{2\mathrm{D}}) = \int_{x_{\mathrm{left}}}^{x_{\mathrm{right}}} 2y_{\mathrm{top}}(x)dx, \quad \Delta\mathcal{A} = \frac{\pi - \mathrm{Area}(\Omega_{2\mathrm{D}})}{\pi} \times 100\%. \tag{11}$$

2. **4D Hypervolume in $\mathbb{H}$:** Using the spherical volume element on the imaginary subspace $d^4q = dx \cdot 4\pi y^2 dy$, the 4D hypervolume of the unit ball $\mathbb{B}_1$ is $V_4(\mathbb{B}_1) = \frac{\pi^2}{2} \approx 4.9348$. The 4D hypervolume of Ω is:

$$V_4(\Omega) = \frac{4\pi}{3} \int_{x_{\text{left}}}^{x_{\text{right}}} (y_{\text{top}}(x))^3 \, dx, \quad \Delta\mathcal{V}_4 = \frac{V_4(\mathbb{B}_1) - V_4(\Omega)}{V_4(\mathbb{B}_1)} \times 100\%. \tag{12}$$

3 Mechanics and Asymptotic Cap Shrinkage

3.1 Matrix Structure of H and K

Proposition 3.1 (**Entries of Cartesian Components**). *Let $C \in M_n(\mathbb{H})$ be the companion matrix (3), indexed with rows and columns $0 \le i, j \le n-1$. The entries of H and K are given by:*

1. ***Hermitian Component H:***

   - *Diagonal: $H_{ii} = 0$ for $0 \le i \le n-2$, and $H_{n-1,n-1} = -\operatorname{Re}(a_{n-1}) \in \mathbb{R}$.*
   - *Intermediate off-diagonals: $H_{i,i+1} = H_{i+1,i} = 1/2$ for $0 \le i \le n-3$, with all other intermediate entries zero.*
   - *Terminal row and column:*

$$H_{n-1,j} = \begin{cases} -\frac{1}{2}a_j \in \mathbb{H}, & 0 \le j \le n-3, \\ \frac{1}{2}(1 - a_{n-2}) \in \mathbb{H}, & j = n-2, \\ -\operatorname{Re}(a_{n-1}) \in \mathbb{R}, & j = n-1, \end{cases} \qquad H_{j,n-1} = (H_{n-1,j})^*. \tag{13}$$

   When all coefficients are real ($a_\nu \in \mathbb{R}$), $H_{n-1,j} = -\frac{1}{2}a_j \le 0$ for $j \le n-3$, and $H_{n-2,n-1} = \frac{1}{2}(1 - a_{n-2}) \ge 0$ since $a_{n-2} \le a_{n-1} = 1$.

2. ***Skew-Hermitian Component K:***

   - *Diagonal: $K_{ii} = 0$ for $0 \le i \le n-2$, and $K_{n-1,n-1} = -\operatorname{Im}(a_{n-1})$.*
   - *Intermediate off-diagonals: $K_{i,i+1} = 1/2$ and $K_{i+1,i} = -1/2$ for $0 \le i \le n-3$.*
   - *Terminal row and column: $K_{n-1,j} = -\frac{1}{2}a_j$ and $K_{j,n-1} = \frac{1}{2}a_j^*$ for $j \le n-3$, with $K_{n-1,n-2} = -\frac{1}{2}(1 + a_{n-2})$.*

3.2 Analysis of the Canonical Degree 2 Case

For degree $n = 2$ with real monotonic coefficients $0 < a_0 \le a_1 \le 1$:

$$C = \begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix} \implies H = \begin{pmatrix} 0 & \frac{1-a_0}{2} \\ \frac{1-a_0}{2} & -a_1 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & \frac{1+a_0}{2} \\ -\frac{1+a_0}{2} & 0 \end{pmatrix}.$$

The eigenvalues of H and singular value of K evaluate analytically to:

$$\lambda_{\max}(H) = \frac{-a_1 + \sqrt{a_1^2 + (1-a_0)^2}}{2}, \quad \sigma_{\max}(K) = \frac{1 + a_0}{2}. \tag{14}$$

Example 3.2 (**Exact Geometric Breakdown for $a_0 = 0.2, a_1 = 0.8$**). Plugging in $a_0 = 0.2$ and $a_1 = 0.8$:

$$\lambda_{\max}(H) \approx 0.1657, \quad \lambda_{\min}(H) \approx -0.9657, \quad \sigma_{\max}(K) = 0.6000.$$

1. **Real Cut Contribution:** The right boundary $x \le 0.1657$ chops off a circular cap of area:

$$A_{\text{real cap}} = \arccos(0.1657) - 0.1657\sqrt{1 - 0.1657^2} \approx 1.2409 \quad (39.50\% \text{ of disk area}).$$

2. **Imaginary Cut Contribution:** The horizontal boundary $|y| \leq 0.6000$ cuts off the top and bottom circular segments for $|y| \in [0.60, 1.0]$.

3. **Combined Reductions:** The total 2D cross-sectional area of Ω shrinks to ≈ 1.3104 (a 58.29% 2D reduction). The 4D hypervolume drops to $V_4(\Omega) \approx 0.9486$ compared to $V_4(\mathbb{B}_1) \approx 4.9348$, giving an 80.78% 4D hypervolume reduction.

3.3 Cauchy Interlacing and Asymptotic Cap Shrinkage

Theorem 3.3 (**Cauchy Interlacing Spectral Lower Bounds**). *For any monic quaternionic polynomial $T(q)$ of degree $n \geq 2$, regardless of coefficient choices, the extreme eigenvalues of the Cartesian components $H = \frac{1}{2}(C + C^*)$ and $K = \frac{1}{2}(C - C^*)$ satisfy:*

$$\lambda_{\max}(H) \geq \cos\left(\frac{\pi}{n}\right), \quad \lambda_{\min}(H) \leq -\cos\left(\frac{\pi}{n}\right), \tag{15}$$

and

$$\sigma_{\max}(K) \geq \cos\left(\frac{\pi}{n}\right). \tag{16}$$

Proof. Look at the complex embedding $\chi(H) \in M_{2n}(\mathbb{C})$. The leading $(n-1) \times (n-1)$ block of H, which I denote by T_{n-1}, is an unmoving, real symmetric tridiagonal Toeplitz matrix with diagonal 0 and off-diagonal 1/2. Its eigenvalues are $\mu_k = \cos(k\pi/n)$ for $k = 1, \ldots, n-1$. Under χ, the submatrix $\chi(T_{n-1}) \in M_{2(n-1)}(\mathbb{C})$ has the exact same eigenvalues, each with multiplicity 2. Applying Cauchy's interlacing theorem to the principal submatrix $\chi(T_{n-1})$ of the Hermitian matrix $\chi(H)$ immediately gives:

$$\lambda_{\max}(H) \geq \mu_1(T_{n-1}) = \cos\left(\frac{\pi}{n}\right), \quad \lambda_{\min}(H) \leq \mu_{n-1}(T_{n-1}) = -\cos\left(\frac{\pi}{n}\right).$$

Similarly, the leading $(n-1) \times (n-1)$ submatrix of K, denoted S_{n-1}, is a skew-symmetric tridiagonal Toeplitz matrix with singular values $|\cos(k\pi/n)|$. Applying Cauchy interlacing for singular values of principal submatrices yields $\sigma_{\max}(K) \geq \sigma_{\max}(S_{n-1}) = \cos(\pi/n)$. □

Theorem 3.4 (**Asymptotic Cap and Collar Shrinkage Upper Bound**). *Let $T(q)$ be a monic polynomial with positive monotonic real coefficients ($0 < a_0 \leq a_1 \leq \cdots \leq a_n = 1$) of degree $n \geq 3$. The total 2D area of the unit disk removable by the Cartesian bounding box is bounded above by four circular caps:*

$$\Delta\,\text{Area} \leq 4A_{cap}(n), \quad \text{where } A_{cap}(n) = \frac{\pi}{n} - \frac{1}{2}\sin\left(\frac{2\pi}{n}\right) = \frac{2\pi^3}{3n^3} + \mathcal{O}(n^{-5}). \tag{17}$$

Consequently, the fractional 2D area reduction is bounded above by:

$$\frac{\Delta\,\text{Area}}{\pi} \leq \frac{4A_{cap}(n)}{\pi} = \frac{8\pi^2}{3n^3} + \mathcal{O}(n^{-5}) \xrightarrow{n\to\infty} 0. \tag{18}$$

Furthermore, in 4D, the real spectral cuts remove at most two polar caps, each of volume:

$$V_{polar}(n) = \frac{4\pi}{3}\int_{\cos(\pi/n)}^{1}(1-x^2)^{3/2}dx = \frac{4\pi^6}{15n^5} + \mathcal{O}(n^{-7}), \tag{19}$$

while the imaginary spectral cutoff $|\operatorname{Im}(q)| \leq \sigma_{\max}(K)$ removes at most an equatorial collar on the 3D imaginary subspace of volume:

$$V_{collar}(n) = \int_{\cos(\pi/n)}^{1} 4\pi y^2 \cdot 2\sqrt{1-y^2}dy = \pi\left[\frac{\pi}{n} - \frac{1}{4}\sin\left(\frac{4\pi}{n}\right)\right] = \frac{8\pi^4}{3n^3} + \mathcal{O}(n^{-5}). \tag{20}$$

Because the equatorial collar dominates the polar caps, the fractional 4D hypervolume reduction is bounded above by:

$$\frac{\Delta V_4}{V_4(\mathbb{B}_1)} \leq \frac{V_{collar}(n) + 2V_{polar}(n)}{V_4(\mathbb{B}_1)} = \frac{16\pi^2}{3n^3} + \mathcal{O}(n^{-5}) \xrightarrow{n\to\infty} 0. \tag{21}$$

Proof. From Theorem 3.3, we know $\lambda_{\max}(H) \geq \cos(\pi/n)$ and $\lambda_{\min}(H) \leq -\cos(\pi/n)$. The real cut enforces $\lambda_{\min}(H) \leq x \leq \lambda_{\max}(H)$, so the portion of the unit disk removed on the right is at most the cap $x \geq \cos(\pi/n)$, and on the left at most $x \leq -\cos(\pi/n)$. Each 2D cap has area $A_{\text{cap}}(n) = \theta_0 - \sin\theta_0 \cos\theta_0$ with $\theta_0 = \pi/n$. Symmetrically, since $\sigma_{\max}(K) \geq \cos(\pi/n)$, the horizontal boundaries $|y| \leq \sigma_{\max}(K)$ cut away at most two circular caps, each of area at most $A_{\text{cap}}(n)$. Adding all four caps together gives at most $4A_{\text{cap}}(n)$. Expanding $\sin(2\pi/n)$ as a Taylor series gives $A_{\text{cap}}(n) = \frac{2\pi^3}{3n^3} + \mathcal{O}(n^{-5})$, establishing (18).

In 4D, the unit ball $\mathbb{B}_1 = \{(x, \mathbf{y}) \in \mathbb{R} \times \mathbb{R}^3 : x^2 + \|\mathbf{y}\|^2 \leq 1\}$ has hypervolume $V_4(\mathbb{B}_1) = \frac{\pi^2}{2}$. The real cuts $|x| \geq \cos(\pi/n)$ define two spherical polar caps. Setting $x = \cos t$ in (19) gives:

$$V_{\text{polar}}(n) = \frac{4\pi}{3} \int_0^{\pi/n} \sin^4(t) dt = \frac{4\pi}{3} \left[\frac{3t}{8} - \frac{\sin(2t)}{4} + \frac{\sin(4t)}{32} \right]_0^{\pi/n} = \frac{4\pi^6}{15n^5} + \mathcal{O}(n^{-7}).$$

Meanwhile, the imaginary cut restricts $\|\mathbf{y}\| \leq \sigma_{\max}(K)$. Since $\sigma_{\max}(K) \geq \cos(\pi/n)$, the removable imaginary region is contained within $\|\mathbf{y}\| \geq \cos(\pi/n)$. Integrating over the radial imaginary coordinate $y = \|\mathbf{y}\| \in [\cos(\pi/n), 1]$ with spherical volume element $4\pi y^2 dy$ and real thickness $2\sqrt{1-y^2}$, and substituting $y = \cos\theta$ ($\theta \in [0, \pi/n]$):

$$V_{\text{collar}}(n) = 8\pi \int_0^{\pi/n} \cos^2\theta \sin^2\theta d\theta = 2\pi \int_0^{\pi/n} \sin^2(2\theta) d\theta = \pi \left[\frac{\pi}{n} - \frac{\sin(4\pi/n)}{4} \right].$$

Expanding $\sin(4\pi/n) = \frac{4\pi}{n} - \frac{32\pi^3}{3n^3} + \mathcal{O}(n^{-5})$ yields $V_{\text{collar}}(n) = \frac{8\pi^4}{3n^3} + \mathcal{O}(n^{-5})$. Summing the collar and the two polar caps and dividing by $V_4(\mathbb{B}_1) = \frac{\pi^2}{2}$ yields the 4D upper bound (21). □

Remark 3.5 (The Practical Regime of the Bound). Theorem 3.4 shows that the unscaled Cartesian box degenerates rapidly relative to $\mathbb{B}_1$, decaying at rate $\mathcal{O}(n^{-3})$ across both 2D and 4D spaces (with the 4D decay rate governed by the 3D imaginary equatorial collar rather than polar caps). As shown in Table 1, the bound delivers its strongest practical reductions for low degrees ($n \leq 5$). At $n = 7$, $\lambda_{\max}(H) \approx 0.9951$ approaches the unit boundary, and beyond $n = 7$ (1.0548 at $n = 8$), the empirical mean exceeds 1.0, which naturally terminates active truncation as the enclosure relaxes back into the classical unit ball.

Table 1: Cauchy interlacing lower bound $\cos(\pi/n)$, empirical mean of $\lambda_{\max}(H)$, mean 2D area reduction, and mean 4D hypervolume reduction across degrees $n = 2$ to 10 (500 trials per degree, uniform prior).

Degree n	**Lower Bound** $\cos(\pi/n)$	**Empirical Mean** $\lambda_{\max}(H)$	**Mean 2D Red. (%)**	**Mean 4D R**
2	0.0000	0.1622	60.57%	76.31
3	0.5000	0.5223	24.72%	26.72
4	0.7071	0.7254	10.40%	8.09%
5	0.8090	0.8462	4.42%	2.80%
6	0.8660	0.9345	1.51%	0.81%
7	0.9010	0.9951	0.50%	0.29%
8	0.9239	1.0548	0.08%	0.04%
9	0.9397	1.1004	0.03%	0.03%
10	0.9511	1.1334	0.01%	0.01%

4 Failure Analysis of Alternative Row-Sum Baselines

To clearly explain why I had to abandon row-sum bounds and move to the Cartesian numerical range in the first place, I present an analytical and numerical autopsy of classical row-sum matrix localization theorems (Geršgorin discs and Brauer Cassini ovals) applied to companion matrices.

4.1 Mechanism 1: The Unit-Disk Collapse of Full Unions

Theorem 4.1 (**Unit-Disk Collapse of Unscaled Row-Sum Enclosures**). *Let $T(x)$ be a monic real polynomial of degree $n \geq 2$ ($a_\nu \in \mathbb{R}$). Let $C \in M_n(\mathbb{R})$ be its direct companion matrix. Then:*

1. *For any $n \geq 2$, the union of Geršgorin discs satisfies $\bigcup_{i=0}^{n-1} \mathcal{D}_i(C) \supseteq \mathbb{D}(0,1)$.*
2. *For any $n \geq 3$, the union of all pairwise Brauer Cassini ovals satisfies $\bigcup_{0 \leq j < k \leq n-1} \mathcal{C}_{jk}(C) \supseteq \mathbb{D}(0,1)$.*

Consequently, both unscaled row-sum formulations achieve exactly 0% area reduction inside $\mathbb{D}(0,1)$.

Proof. For the companion matrix C, the intermediate rows $i = 0, 1, \ldots, n-2$ have diagonal entry $c_{ii} = 0$ and deleted row sum $R_i = 1$.

1. Because $c_{ii} = 0 \in \mathbb{R}$ commutes with all eigenvalues, the Geršgorin disc for every intermediate row is $\mathcal{D}_i(C) = \{q \in \mathbb{H} : |q| \leq 1\} = \mathbb{D}(0,1)$. Hence, $\bigcup \mathcal{D}_i \supseteq \mathbb{D}(0,1)$.
2. For $n \geq 3$, there exist at least two distinct intermediate rows $j, k \leq n-2$ with $j < k$. Both centers are 0 and both row sums are 1. The pairwise Cassini oval between them is $\mathcal{C}_{jk}(C) = \{q \in \mathbb{H} : |q|^2 \leq 1\} = \mathbb{D}(0,1)$. Hence, $\bigcup_{j<k} \mathcal{C}_{jk} \supseteq \mathbb{D}(0,1)$.

Intersecting either union with $|q| \leq 1$ yields identically $\mathbb{D}(0,1)$, giving exactly 0% reduction. □

Remark 4.2 (**Diagonal Similarity Scaling**). Unscaled row-sum bounds on companion matrices are notoriously loose. Standard fixes rely on diagonal similarity scaling $\widetilde{C} = D^{-1}CD$ with $D = \mathrm{diag}(d_1, \ldots, d_n)$ to balance row sums (like in Fujiwara and Kojima scaling). Diagonal similarity can also be combined with Cartesian decomposition by evaluating the numerical range of $D^{-1}CD$, which is a promising direction for future work.

4.2 Mechanism 2: Root Leakage in the Single Cassini Oval Heuristic

An intuitive trick to try when attempting to tighten Gershgorin circles is isolating a single Cassini oval between the first row and the terminal row of the Cauchy-shifted companion matrix $C_\phi \in M_{n+1}(\mathbb{R})$ (obtained from $\phi(q) = T(q) * (1 - q)$):

$$\mathcal{B}_{\text{single}} = \{q \in \mathbb{H} : |q||q - c_{nn}| \leq R_n\}, \tag{22}$$

where $c_{nn} = 1 - a_{n-1}$ and $R_n = a_{n-1}$. Isolating a single pair technically violates Brauer's union theorem (since eigenvalue containment is only guaranteed for the union of all pairwise ovals). But one might still wonder if the terminal oval alone empirically manages to hold all roots under monotonic coefficients. The answer is a clear no: it fundamentally leaks zeros.

Theorem 4.3 (**Root Leakage of the Isolated Cassini Oval Heuristic**). *Let $T(x) = \sum_{\nu=0}^{n} a_\nu x^\nu$ be a real polynomial with monotonic positive coefficients $0 < a_0 \leq a_1 \leq \cdots \leq a_n = 1$. Any real negative zero $z = -r$ ($r > 0$) of $T(x)$ lies in the isolated Cassini oval $\mathcal{B}_{single}$ if and only if:*

$$r \leq a_{n-1}. \tag{23}$$

Consequently, every real zero satisfying $a_{n-1} < r \leq 1$ escapes $\mathcal{B}_{single}$.

Proof. By Descartes' rule of signs, any real root must be negative: $z = -r$ with $r > 0$. Substituting $z = -r$ into $|z||z - c_{nn}| \leq R_n$ yields $r(r + c_{nn}) \leq R_n$. The positive root of $r^2 + c_{nn}r - R_n = 0$ is $r = \frac{-c_{nn} + \sqrt{c_{nn}^2 + 4R_n}}{2}$. Since $c_{nn} + R_n = 1$, the discriminant simplifies to $(2 - c_{nn})^2$. Because $c_{nn} \in [0,1]$, $r \leq \frac{-c_{nn} + (2 - c_{nn})}{2} = 1 - c_{nn} = a_{n-1}$. Any zero with $a_{n-1} < r \leq 1$ escapes. □

In my empirical tests across $10,000$ randomly generated monotonic polynomials, the single Cassini oval $\mathcal{B}_{\text{single}}$ failed in $2,367$ out of $10,000$ cases (23.67% failure rate), with escaping roots blowing past the boundary by margins up to -0.6089.

5 Empirical Verification and Adversarial Stress Testing

To verify the numerical stability and practical performance of this framework, I implemented an automated test suite in Python using 64-bit IEEE floating-point arithmetic.

Root Extraction via Eigenvector Conjugation. For quaternionic polynomials, roots were computed from the complex eigenvalues $\lambda \in \mathbb{C}$ of the companion matrix lifted to $\chi(C)$ using the Serôdio-Pereira-Vitória method [3]: for each eigenvector $w = (w_1, \ldots, w_{2n})^T$, the quaternion $x_0 = \text{Re}(w_1) + \text{Im}(w_1)\mathbf{i} - \text{Re}(w_2)\mathbf{j} + \text{Im}(w_2)\mathbf{k}$ is formed, and the true right zero of $T(q)$ is obtained via quaternionic conjugation:

$$q = x_0 \lambda x_0^{-1}. \tag{24}$$

All computed roots were verified by direct left polynomial evaluation, recovering residuals $\|T(q)\| \sim 10^{-14}$ at machine precision.

5.1 Comparative Audit: Cassini Heuristic vs. Cartesian Bounds

Table 2 presents the comparison across $10,000$ Monte Carlo trials across degrees $n \in \{2, 3, 4, 5, 6\}$. Real coefficients were generated by sorting uniform random draws $a_\nu \sim \mathcal{U}(0, 1]$ normalized so that $a_n = 1$. The reported reductions reflect the expected values under this uniform prior.

Table 2: Zero containment rates, boundary slacks, and mean 2D area reductions across $10,000$ trials ($n \in \{2, 3, 4, 5, 6\}$, uniform prior).

Method	**Containment Rate**	**Min. Slack**	**Mean 2D Red.**	
Restricted Real Monotonic Polynomials ($N = 10,000$)				
Classical Unit Disk $\mathbb{B}_1$	100.0%	+0.0056	Reference (0.0%)	V
Full Geršgorin / Brauer Union	100.0%	+0.0056	0.0%	Co
Single Cassini Oval $\mathcal{B}_{\text{single}}$	**76.33%**	**−0.6089**	19.63%	**Fails** (
Proposed Cartesian Ω	**100.0%**	**Box: +0.0861**	**20.32%** (mean deg 2–6)	**St**

5.2 Adversarial Numerical Stress Test Suite ($196,059$ Polynomials)

To test numerical conditioning and stability under extreme coefficient ratios and near-boundary configurations, I designed and ran seven adversarial stress test batteries across $196,059$ polynomials in total. Zero violations were observed under strict tolerance (tol $= 0$).

Sensitivity to Prior Distributions and Slack Margins: The area reductions reported in Table 1 reflect the baseline uniform prior ($a_\nu \sim \mathcal{U}(0, 1]$). Looking at different coefficient regimes:

- **Marginal Monotonicity / Flat Spectra (Battery 3, $n = 3 \ldots 6$):** With $a_k = \frac{1+k\varepsilon}{1+n\varepsilon}$ as $\varepsilon \to 0$, the mean area reduction stabilizes at 16.72%.
- **Steep Exponential Ramps (Battery 4, $n = 3 \ldots 6$):** For dynamic ranges a_n/a_0 spanning up to 10^8, the mean area reduction increases to 27.26%. Across all distributions, the Cartesian box strictly encloses the full spectrum.

Table 3: Summary of the Seven Adversarial Stress Batteries for Numerical Stability (196,059 total evaluated).

Battery	Stress Focus	Polynomials Evaluated	Containment
1	High-degree degeneration ($n = 2 \dots 15$)	7,000	100.0%
2	Near-boundary roots ($\lvert q\rvert \geq 0.90$, 2,000 isolated)	32,059	100.0%
3	Marginal monotonicity ($\varepsilon \to 0$, $n = 3 \dots 6$)	2,500	100.0%
4	Extreme dynamic range ($a_n/a_0 \leq 10^8$, $n = 3 \dots 6$)	2,000	100.0%
5	Large imaginary scale ($\lVert \operatorname{Im} \rVert \leq 2.0$, $n = 3 \dots 6$)	2,500	100.0%
6	Slack distribution census ($n = 3$)	50,000	100.0%
7	Active zero-tolerance violation search	100,000	100.0%
Total	**Exhaustive Adversarial Suite**	**196,059**	**100.0%**

- **Conditioning and Residuals:** At coefficient ratio 10^8, maximum polynomial evaluation residuals $\|T(q)\|$ increased from 6.78×10^{-14} to 6.73×10^{-8}, consistent with standard IEEE 754 precision limits, with zero containment failures.
- **Empirical Slack Census (Battery 6):** Across 50,000 sampled polynomials (150,000 roots), the observed minimum slacks stayed strictly positive (0.1170 for $\operatorname{Re}(q) \geq \lambda_{\min}(H)$, 0.2175 for $\operatorname{Re}(q) \leq \lambda_{\max}(H)$, and 0.0861 for $|\operatorname{Im}(q)| \leq \sigma_{\max}(K)$). This empirical cushion confirms that across typical and near-boundary configurations ($|q| \geq 0.90$), roots do not cluster arbitrarily close to the spectral boundary, preventing numerical leakage under floating-point round-off.

6 Conclusion

In this paper, I built a coordinate-decomposed eigenvalue enclosure framework for the zeros of quaternionic polynomials based on the numerical range of the direct Frobenius companion matrix. By using the Cartesian decomposition $C = H + K$ and Rayleigh quotient bounds over $\mathbb{H}$, I derived bounds on $\operatorname{Re}(q)$ and $|\operatorname{Im}(q)|$ that unconditionally hold for any monic polynomial over $\mathbb{H}$. For real monotonic polynomials, the resulting Cartesian hyper-cylinder truncates the classical Eneström-Kakeya ball, giving massive geometric reductions for low degrees (for $n = 2$, mean reductions of 60.57% in 2D area and 76.31% in 4D hypervolume, and 24.72% in 2D and 26.72% in 4D for $n = 3$). For general quaternionic polynomials with non-real coefficients, the Cartesian cylinder unconditionally encloses the full 4D spectrum.

Using Cauchy's interlacing theorem on the leading Toeplitz submatrices of H and K, I established the spectral lower bounds $\lambda_{\max}(H) \geq \cos(\pi/n)$, $\lambda_{\min}(H) \leq -\cos(\pi/n)$, and $\sigma_{\max}(K) \geq \cos(\pi/n)$, proving that the maximum removable fractional area is bounded by $\frac{8\pi^2}{3n^3} + \mathcal{O}(n^{-5})$ in 2D and the fractional hypervolume by $\frac{16\pi^2}{3n^3} + \mathcal{O}(n^{-5})$ in 4D (governed by the 3D imaginary equatorial collar). This establishes a rigorous asymptotic upper bound decaying at rate $\mathcal{O}(n^{-3})$ across all dimensions, while the empirical growth of $\lambda_{\max}(H) > 1$ at $n \geq 8$ explains the rapid practical termination of active truncation beyond low degrees. Furthermore, I documented why alternative row-sum heuristics fail: unscaled Geršgorin and Brauer unions collapse to the unit disk (0% reduction for $n \geq 3$), while isolating a single Cassini oval leaks roots (23.67% failure rate). Extensive empirical testing across 196,059 polynomials confirmed numerical stability under extreme conditions. A natural direction for future work is combining Cartesian decomposition with optimal diagonal similarity scaling $D^{-1}CD$ to further tighten the enclosure across higher degrees.

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